Scalability & Future Plan

Date	15 March 2026
Team ID	XXXXXX
Project Name	Electric Vehicle Market Performance & Insights
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Limited dataset size and availability of updated EV market data.	May affect the accuracy of future market analysis.	High
2	Dashboard performance may decrease with larger datasets.	Slower loading and filtering of visualizations.	Medium
3	Limited user access and security features in the current Flask website.	Not suitable for large-scale public usage.	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Designed as an individual project for demonstration.	Deploy on a scalable cloud hosting platform to support multiple users.
2	Data Storage	Uses the existing EV dataset containing 200,000 rows and 38 columns.	Integrate a database to store and manage larger EV datasets.
3	Performance	Tableau dashboards and Flask website display project visualizations.	Optimize data processing and dashboard loading for larger datasets.
4	Security	Basic Flask website setup.	Implement user authentication and secure access controls.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Integration of updated and real-time EV market data.	Future Enhancement	Provides more accurate and updated market insights.
Phase 3	Addition of advanced analytics and EV sales prediction.	Future Enhancement	Helps identify future EV market trends.
Phase 4	Cloud deployment and improved website functionality.	Future Enhancement	Makes the project more scalable and accessible to multiple users.